Digital Signal Processing

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- Why study FFT algorithm
- The historical evolution of FFT algorithm
- Basic ideas for solving problems

Fast algorithm for discrete Fourier transform

$$x(t) \xrightarrow[\text{FT}]{\text{theoretically}} X(\mathrm{j}\omega)$$

$$x[k] \xrightarrow[\text{FT}]{\text{theoretically}} X(\mathrm{e}^{\mathrm{j}\Omega})$$

$$x[k] \xrightarrow[\text{DFT}]{\text{methodologically}} X[m]$$

The computational complexity of the 4-point sequence {2,3,3,2} DFT

$$X[m]=\sum_{k=0}^{N-1}x[k]W_N^{km},\quad m=0,1,\cdots,N-1 \qquad W_N^{km}=\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}km}$$

$$X[0]=2W_N^0+3W_N^0+3W_N^0+2W_N^0=10$$

$$X[1]=2W_N^0+3W_N^1+3W_N^2+2W_N^3=-1-\mathrm{j}$$

$$X[2]=2W_N^0+3W_N^2+3W_N^4+2W_N^6=0$$

$$X[3]=2W_N^0+3W_N^3+3W_N^6+2W_N^9=-1+\mathrm{j}$$

Complex addition $N(N-1)$

Complex multiplication N^2

As N increases, the computational complexity of DFT increases sharply

The importance of DFT in digital signal processing has long been recognized, but its widespread application has taken a long time due to its high computational complexity and difficulty in practical use.

Since the mid-20th century, significant progress has been made in the research of fast algorithms for signal processing, including the fast algorithm for DFT, namely Fast Fourier Transform (FFT), which is a general term for various DFT fast algorithms.

Fast algorithm for discrete Fourier transform

$$x(t) \xrightarrow[\text{FT}]{\text{Theoretically}} X(\mathrm{j}\omega)$$

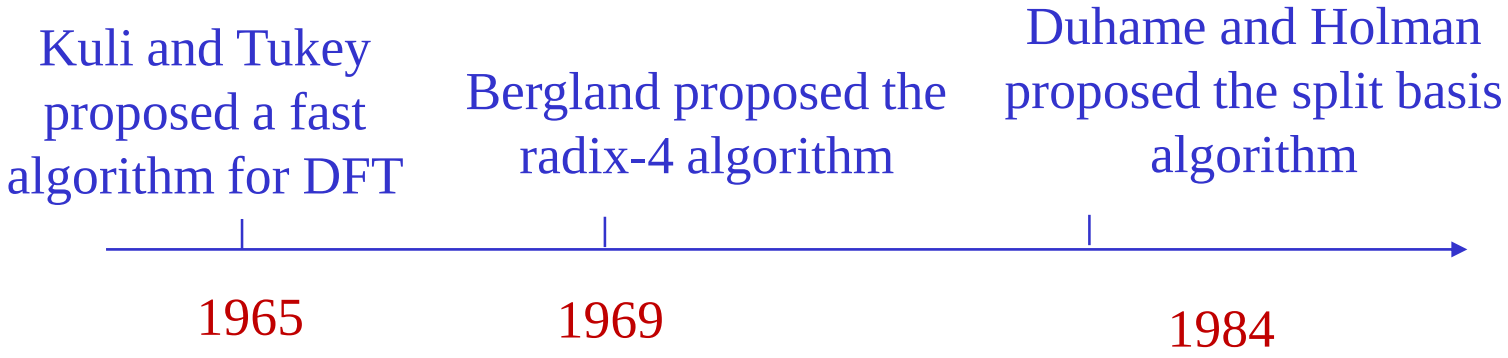
$$x[k] \xrightarrow{\qquad} X(\mathrm{e}^{\mathrm{j}\Omega})$$

$$x[k] \xrightarrow[\text{DFT}]{\text{Methodologically}} X[m]$$

$$x[k] \xrightarrow[\text{FFT}]{\text{Technically}} X[m]$$

The historical evolution of FFT algorithm

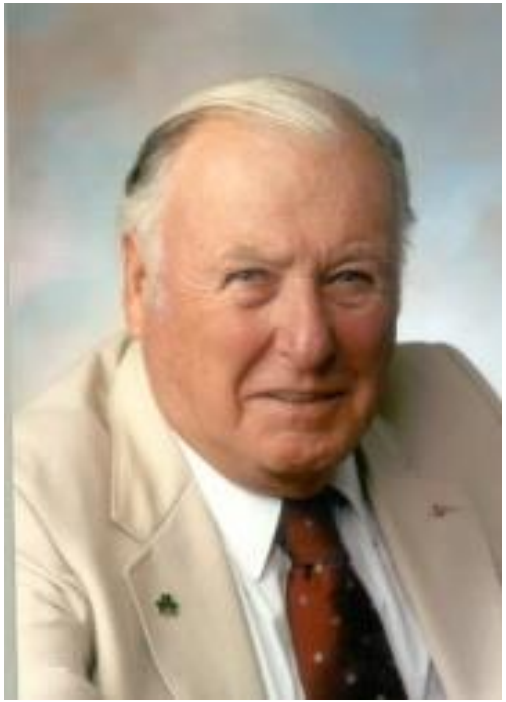
In 1965, Cooley and Tukey's fast algorithm underwent several improvements and formed an efficient computational method, which is now known as FFT. This algorithm greatly improves the computational efficiency of DFT and reduces the cumulative quantization error caused by DFT operations.



Fast algorithm for discrete Fourier transform

Kuli, James William Cooley(1926-)， The Fast Fourier Transform (FFT) proposed by American mathematician and Ph.D. in Mathematics from Columbia University has made breakthrough progress in digital signal processing technology, laying the foundation for the development of fields such as network communication and graphics and image processing.

For many years, I have been mainly engaged in research on digital signal processing at IBM Research Center. In 1980, I received the ASSP Society and Meritorious Service Award, and in 1984, I received the Society Award and IEEE Centennial Medal.



Fast algorithm for discrete Fourier transform



Tuji, John Wilder Tukey (1915~2000)，A renowned American mathematician with a PhD in Mathematics from Princeton University, he and Cooley jointly proposed a fast Fourier transform algorithm.

In 1973, he was awarded the National Medal of Science in the United States, and in 1982, he was awarded the IEEE Medal of Honor and a Foreign Fellow of the Royal Academy of Sciences. The Tukey Range Test, Tukey Lambda Distribution, Tukey Additive Test, and Tukey Lemma, all named after him, were his lifelong scientific contributions.

Basic ideas for solving problems

The basic idea of the FFT algorithm proposed by Cooley Tukey is:

※ Decompose long sequence DFT into short sequence DFT

※ Utilize the periodicity, symmetry, and reducibility of rotation factors

Basic ideas for solving problems

The properties of rotation factor W_N^{km}

$$W_N = e^{-j\frac{2\pi}{N}}$$

(1) periodicity $W_N^{(k+N)m} = W_N^{k(m+N)} = W_N^{km}$

(2) symmetry $W_N^{mk+\frac{N}{2}} = -W_N^{mk}$ $\left(W_N^{km}\right)^* = W_N^{-mk}$

(3) Reducibility $W_N^{mk} = W_{MN}^{Mmk}$ $W_N^{mk} = W_{N/M}^{mk/M}$, N/M is integer

Basic ideas for solving problems

Decompose the time-domain sequence into a set of subsequences, and utilize the characteristics of rotation factors to achieve the DFT of the entire sequence through the DFT of the subsequences.

※ Decimation in time (DIT) FFT algorithm

$$x[k] \rightarrow \begin{cases} x_1[k] = x[2k] \\ x_2[k] = x[2k+1] \end{cases} \qquad k = 0, 1, \cdots, \frac{N}{2} - 1$$

※ Decimation in Frequency (DIF) FFT Algorithm

$$x[k] \rightarrow \begin{cases} x_1[k] = \{x[k] + x[k + N/2]\} W_N^0 \\ x_2[k] = \{x[k] - x[k + N/2]\} W_N^k \end{cases} \qquad k = 0, 1, \cdots, \frac{N}{2} - 1$$

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.